# OpenScript Studio — Comprehensive QA & Functional Testing Guide

**Local Studio URL:** http://localhost:5173
**Backend API URL:** http://localhost:3000
**Target Environments:** Dev Preview (`dev-openscript.ssnpictures.in`) & Production (`openscript.ssnpictures.in`)

## Overview

This guide provides test cases and scenarios to systematically verify every button, keyboard trigger, outline tool, revision workflow, export format, and collaboration control in OpenScript Studio.

## Test Suite 1: Screenplay Canvas & Deterministic Keyboard Engine

### Test 1.1: Element State Machine (`Tab` and `Enter`)

- **Objective:** Verify deterministic cycling between screenplay elements without using the mouse.
- **Steps:**
  1. Open http://localhost:5173 in Screenplay mode.
  2. Click into the editor and press `Enter` on a blank line.
  3. Type `INT. ABANDONED WAREHOUSE - NIGHT` → Press `Enter`.
     - **Expected:** Next line automatically converts to **Action** block.
  4. Type an action line: `Rain drips through the rusted corrugated roof.` → Press `Enter`.
     - **Expected:** Remains **Action**.
  5. Press `Tab`.
     - **Expected:** Block converts to **Character** (centered at 3.7", auto-capitalized).
  6. Type `MAYA` → Press `Enter`.

    - **Expected:** Next line automatically converts to **Dialogue** (indented at 2.5").
7. Type `We need to move before security notices.` → Press `Enter`.
    - **Expected:** Returns to **Action** block.
8. From a blank Action line, press `Shift-Tab`.
    - **Expected:** Block converts to a **Transition** (aligned at 6.0", e.g., `CUT TO:`).

### Test 1.2: Predictive SmartType Autocomplete

- **Steps:**
    1. In an Action block, press `Tab` to create a Character block.
    2. Type the letter `M`.
        - **Expected:** Floating amber suggestion pill appears displaying `MAYA`.
    3. Press `Tab` or click the pill.
        - **Expected:** Automatically completes the character name to `MAYA`.
    4. On a new line, type `IN`.
        - **Expected:** Suggestion pill offers `INT.`. Pressing `Tab` inserts `INT.`.

### Test 1.3: Simultaneous Dual Dialogue Insertion

- **Steps:**
    1. Click into any scene in the editor canvas.
    2. Look for the floating block controls or insert dual dialogue block.
    3. **Expected:** Generates two parallel side-by-side columns:
        - Left column: `CHARACTER 1` + speech.
        - Right column: `CHARACTER 2` + overlapping speech.

## Test Suite 2: Writer Ergonomics & Productivity Suite

### Test 2.1: Typewriter Scrolling Mode Button

- **Location:** Top Ribbon → → `Typewriter` button.
- **Steps:**
    1. Click the **Typewriter** button on the top ribbon.

        - **Expected:** The button illuminates with an amber highlight (`bg-amber-500/20`).
    2. Scroll down and type several new paragraphs.
        - **Expected:** The viewport smoothly autoscrolls to keep the active text cursor locked at approximately 45% screen height, eliminating neck strain.
    3. Click **Typewriter** again to toggle off.

### Test 2.2: Writing Sprint Stopwatch Widget

- **Location:** Top Ribbon → → `Sprint` button.
- **Steps:**
    1. Click the **Sprint** button.
        - **Expected:** The Sprint Stopwatch modal opens in the bottom-right corner.
    2. Click preset `15m` or `25m`.
    3. Enter a target goal (e.g., `200` words).
    4. Click **Start Sprint**.
        - **Expected:** Timer starts counting down; the live progress bar dynamically tracks words typed during the active session.
    5. Click **Pause** or **Reset** to verify timer controls.

### Test 2.3: Dialogue Voice Isolator

- **Location:** Top Ribbon → → `Voice: All` dropdown.
- **Steps:**
    1. Click the **Voice** dropdown in the header.
    2. Select `MAYA`.
        - **Expected:** All non-Maya dialogue lines, scene headings, and action blocks dim to 25% opacity with grayscale filter.
        - **Expected:** All dialogue spoken by `MAYA` remains 100% bright with a subtle warm highlight, allowing you to proofread her rhythm across the entire draft.
    3. Select `Voice: All` to restore normal full-contrast display.

### Test 2.4: Focus Mode (Distraction-Free)

- **Location:** Left Sidebar Footer → → `Focus Mode` toggle.
- **Steps:**

1. Click **Focus Mode**.
    - **Expected:** The left Scene Navigator sidebar collapses to `0px`, centering the paper canvas with zero distractions.
2. Click the exit focus icon to restore the studio layout.

## Test Suite 3: Hollywood Production Revision Tracking & Script Locking

### Test 3.1: Hollywood 9-Color Revision Hierarchy

- **Location:** Top Ribbon → → Revision Color Dropdown (Default: `White Production Draft`).
- **Steps:**
    1. Click the revision dropdown button showing the color dot.
        - **Expected:** A menu opens listing standard Hollywood production colors:
            - White → → Blue → → Pink → → Yellow → → Green → → Goldenrod → → Buff → → Salmon → → Cherry.
    2. Click `Blue Draft (1st Rev.)`.
        - **Expected:** The ribbon button changes to blue with hex `#60a5fa`.
        - **Expected:** The Title Page metadata and page headers update to reflect the "Blue Production Draft".

### Test 3.2: Right-Margin Revision Asterisks (`*`)

- **Location:** Top Ribbon → → `* Track` button.
- **Steps:**
    1. Click the `* Track` button.
        - **Expected:** Button highlights in red indicating revision tracking is active.
    2. Edit any sentence in an Action block or Dialogue line.
        - **Expected:** A red Hollywood revision asterisk (`*`) immediately renders in the far right margin of that line (`.revision-marker::after`).
    3. Click `* Track` again to disable tracking.

### Test 3.3: Locked Script Scene Numbering Engine

- **Location:** Top Ribbon → → 🔓 `Lock` button.

- **Steps:**
    1. Note the existing scene numbers: Scene 1, Scene 2.
    2. Click the `Lock` button.
        - **Expected:** Button switches to amber `🔒 Locked`.
    3. Insert a new Scene Heading between Scene 1 and Scene 2: `INT. BASEMENT - NIGHT`.
        - **Expected:** The new insert is assigned scene number `1A` (Hollywood insert rule) without changing Scene 2 to Scene 3.

## Test Suite 4: Story Outlining, Navigator & Beat Board

### Test 4.1: Left Sidebar Scene Navigator

- **Location:** Left Sidebar → → Outliner tab.
- **Steps:**
    1. Verify all scenes in the current script are listed with:
        - Scene number badge
        - Slugline (e.g., `INT. COFFEE SHOP - DAY`)
        - Duration in eighths (e.g., `1 2/8 pgs`)
        - Character badges present in the scene
    2. Click `+ Add Scene` in the sidebar.
        - **Expected:** Appends a new scene block into the editor canvas and scrolls to it.
    3. Click the filter pills: `INT` or `EXT`.
        - **Expected:** Filters the list to show only interior or exterior scenes.
    4. Type a keyword in the "Search scenes..." input.
        - **Expected:** Live filters scenes matching slugline or action snippet.
    5. Click on any scene card in the navigator.
        - **Expected:** Smoothly scrolls the screenplay canvas directly to that scene heading.

### Test 4.2: Visual Beat Board (Corkboard / Index Cards)

- **Location:** Top-Left Mode Selector → → `Beats` tab.
- **Steps:**

1. Click `Beats` in the top navigation bar.
    - **Expected:** View switches to the visual corkboard layout.
2. Inspect the index cards:
    - Color-coded header by INT (Warm Amber) / EXT (Sky Blue).
    - Action snippet preview.
    - Speaking character pills.
3. **Drag and Drop Reordering:**
    - Click and drag Index Card #2 above Index Card #1.
    - Release the mouse.
    - **Expected:** Cards visually swap positions.
4. Switch back to `Screenplay` mode.
    - **Expected:** The underlying script AST has reordered the scene sequence to match the board.

## Test Suite 5: Automated Production Reports

### Test 5.1: Speaking Cast Breakdown Report

- **Location:** Top-Left Mode Selector → `Reports` tab → `Speaking Cast`.
- **Steps:**
    1. Click `Reports` in the top navigation.
    2. Click the **Speaking Cast** sub-tab.
        - **Expected:** Displays every speaking character with:
            - Total scenes appeared in
            - Total dialogue blocks
            - Total spoken words
            - Percentage of total script dialogue (progress bar)
            - Character list sorted by volume of dialogue.

### Test 5.2: Locations & Sets Breakdown

- **Location:** `Reports` tab → `Locations`.
- **Steps:**

1. Click the **Locations** sub-tab.
    - **Expected:** Displays:
        - Breakdown cards for unique sets (e.g., `COFFEE SHOP`, `ALLEYWAY`).
        - Setting badge (`INT` vs `EXT`).
        - List of scenes occurring in that location.

### Test 5.3: Day-Out-of-Days (DOOD) Matrix

- **Location:** `Reports` tab → → `DOOD Schedule`.
- **Steps:**
    1. Click the **DOOD Schedule** sub-tab.
        - **Expected:** Renders the standard Hollywood production scheduling matrix:
            - Columns = Scenes (Scene 1, Scene 2, Scene 3...).
            - Rows = Characters.
            - Cell Badges:
                - `SW` = Start Work
                - `W` = Work
                - `H` = Hold
                - `FW` = Finish Work
                - `SWF` = Start-Work-Finish (single scene appearance)

### Test 5.4: Pacing & Scene Eighths Analysis

- **Location:** `Reports` tab → → `Pacing Analysis`.
- **Steps:**
    1. Click the **Pacing Analysis** sub-tab.
        - **Expected:** Renders Action vs. Dialogue percentage ratio bar and average scene duration.

## Test Suite 6: Industry Title Page, Security Watermarking & Studio Exports

### Test 6.1: Title Page & Watermarking Generator

- **Location:** Left Sidebar Footer → → `Title Page` button.
- **Steps:**
    1. Click the **Title Page** button.
        - **Expected:** Opens the Title Page & Security Watermark configuration modal.
    2. Edit:
        - Title: `THE GLASS COUNTER`
        - Author: `Gourav Siddoju`
        - Watermark Text: `CONFIDENTIAL - SSN PICTURES`
        - Check **"Show Security Watermark across all pages"**.
    3. Click **Apply Changes**.
        - **Expected:** A translucent diagonal security watermark appears across the paper canvas and print preview.

### Test 6.2: Final Draft XML ( `.fdx` ) Export & Import

- **Steps:**
    1. Click `Export FDX` in the top ribbon.
        - **Expected:** Downloads a valid `.fdx` XML file ( `THE_GLASS_COUNTER.fdx` ) preserving scene numbers, characters, parentheticals, and dialogue.
    2. Click `Import FDX` in the top ribbon.
    3. Select any `.fdx` file on your computer.
        - **Expected:** Studio canvas immediately parses the XML and populates the screenplay canvas.

### Test 6.3: Standard Fountain ( `.fountain` ) Export

- **Steps:**
    1. Click the `Fountain` button in the top ribbon.

- **Expected:** Downloads a clean plain-text `.fountain` file with proper sluglines and transitions.

### Test 6.4: Print & Studio PDF Export

- **Steps:**
  1. Click the amber `PDF` button in the top ribbon.
     - **Expected:** Launches the browser print preview formatted under `@media print`:
       - Cover Title Page as Page 1.
       - Script body starts at Page 1 (running header `1.` top right).
       - Courier Prime 12pt layout with side margins intact.

## Test Suite 7: Realtime Collaboration, ScriptNotes & Roles (Phase 3)

### Test 7.1: Active Collaborators Presence Bar

- **Location:** Top-Right Header → → Collaborator Avatars.
- **Steps:**
  1. Look at the top-right header next to the `Share` button.
     - **Expected:** Shows active collaborator avatars (e.g., `G`, `N`, `S`) with green pulsing live indicators.
  2. Hover over any avatar:
     - **Expected:** Tooltip shows collaborator display name, role (`owner` / `co-writer`), and live typing state.

### Test 7.2: ScriptNotes & Threaded Feedback Panel

- **Location:** Top-Right Header → → `Notes` button.
- **Steps:**
  1. Click the `Notes` button in the header.
     - **Expected:** The ScriptNotes drawer slides in from the right.
  2. **Create a Note:**
     - Click `Leave ScriptNote on Script`.
     - Select category: `Director`.

- Type content: `Check lighting on the trench coat in the alley.`
- Click **Post Note**.
- **Expected:** Note appears at the top of the list and saves to backend (`/api/v1/documents/:id/notes`).

3. **Reply to Note Thread:**
   - On any note card, type a reply in the "Reply to thread..." box: `Permits are confirmed for night shoot.` → → Press `Enter`.
   - **Expected:** Threaded reply block appears immediately with author avatar and timestamp.
4. **Resolve Note:**
   - Click the checkmark icon (✓✓) on the note card.
   - **Expected:** Note is struck through with green resolved state; unread counter on the top ribbon decreases.
5. **Category Filters:**
   - Click the category dropdown → → select `Dialogue`.
   - **Expected:** Filters the notes list to only dialogue feedback.

### Test 7.3: Team Sharing & Granular Role Simulator

- **Location:** Top-Right Header → → `👥 Share` button.
- **Steps:**
  1. Click `Share`.
     - **Expected:** Opens the "Share Screenplay & Team Access" modal.
  2. **Invite Collaborator:**
     - Enter email: `editor@hollywood.com`.
     - Role: `Co-Writer`.
     - Click **Invite**.
     - **Expected:** Added to the project member list and saved to backend (`/api/v1/projects/:id/collaborators`).
  3. **Copy Invite Link:**
     - Click `Copy Link`.
     - **Expected:** Button turns green showing `Copied!` and copies room URL to clipboard.
  4. **Test Reader Mode (Permission Enforcement):**

- Under "Live Role Simulator", click `Reader`.
- Close the modal.
- **Expected:**
  - A red banner appears above the canvas: `Viewing in Read-Only Mode (Reader Role) — Editing and formatting controls are disabled.`
  - Typing inside the editor is completely blocked.
- Open **Share** → → Click `Owner` or `Co-Writer` to unlock editing.

## Test Suite 8: Backend Autosave & API Health

### Test 8.1: Resilient Autosave Status

- **Location:** Bottom Studio Footer.
- **Steps:**
  1. Type a word anywhere in the screenplay.
     - **Expected:** Footer immediately updates to `Saving...` with amber clock.
  2. Stop typing for 1.5 seconds.
     - **Expected:** Footer flips to green `✓ Saved locally`, having dispatched `PATCH /api/v1/documents/:id` to the Fastify backend.

### Test 8.2: Backend Discovery Endpoint

- **Steps:**
  1. Open a new browser tab to http://localhost:3000.
     - **Expected:** Returns status 200 with service information and direct endpoint links:

```
json
{
  "service": "OpenScript Studio API",
  "version": "0.2.0",
  "status": "online",
  "studioUiUrl": "http://localhost:5173"
}
```

  2. Open http://localhost:3000/api/v1/health.

- **Expected:** Returns `{"status":"ok","service":"openscript-api"}`.

## Quick Testing Results Log

| Suite # | Test Scenario | Verified Status |
|---|---|---|
| **1** | Element State Machine ( `Tab` / `Enter` / Autocomplete) | [ ] Pass / [ ] Fail |
| **2** | Ergonomics (Sprint Stopwatch, Typewriter, Voice Isolator) | [ ] Pass / [ ] Fail |
| **3** | Revisions (Hollywood 9 Colors, Asterisks `*`, Locked Scenes `1A`) | [ ] Pass / [ ] Fail |
| **4** | Outlining (Scene Navigator, Drag & Drop Beat Board) | [ ] Pass / [ ] Fail |
| **5** | Production Reports (Cast Speaking, Locations, DOOD Grid) | [ ] Pass / [ ] Fail |
| **6** | Title Page, Watermark, FDX Import/Export, PDF Export | [ ] Pass / [ ] Fail |
| **7** | Realtime Collaboration (Presence Bar, ScriptNotes, Role Simulator) | [ ] Pass / [ ] Fail |
| **8** | Backend Autosave & API Discovery Endpoints | [ ] Pass / [ ] Fail |